

Stimulate Synergistic Efficiency of Pollution-Carbon Co-Control in the Digital Era: The Role of Digital Trade in Cities

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Abstract

As micro-nodes in the global economic network, cities are hubs of both human activity and industrial production. Cities act as the main **space** of ecological pollution while hosting the most pollution-exposed population, which is a structural paradox that continues to challenge green urban development. The ongoing **digital transformation** of urban industrial structures and **daily** practices, presents new opportunities for cities to pursue more sustainable development paradigms. This study focuses on the role of digital trade in enhancing the synergistic efficiency of pollution control and carbon reduction via panel data covering 283 Chinese cities. **The empirical results indicate** that the leap of digital trade significantly advances the synergistic efficiency of pollution-carbon co-control, with an improvement equivalent to about 20% of the variation range in the synergistic effect. Mechanism analysis reveals dual transmission channels—regional innovation capacity strengthening (contributing 13.33% of total effects) and factor market distortion alleviation (accounting for 24%). The heterogeneity effect exists across cities, and digital trade shows larger green and low-carbon dividends in free trade zones, industrially developed cities and large-scale cities. Crucially, panel threshold regression identifies the optimal level of digital trade for green, low-carbon development (threshold=0.139), beyond which energy-intensive scale-up offsets 29.27% of the initial environmental gains—a phenomenon known as the green efficiency paradox. Regional ecological governance should leverage digital trade to support green transition, strategies should align with local industrial base. **Corresponding policies** should be placed on fostering innovation and optimizing factor market to maximize environmental gains.

Keywords: Digital trade; Synergistic control; Carbon emission; Regional ecological governance; Pollution reduction; Innovation capabilities

1. Introduction

The urbanization of human society and simultaneous industrialization have emerged as critical **sources** of environmental degradation and carbon emissions (Zhao & Lee, 2024; Lee & Wen, 2025). Accelerating urban-industrial agglomeration exerts enormous pressure on micro eco-environment quality (Sha et al., 2025), particularly in developing countries and emerging market economies, where the growth of industrial economy not only causes huge pressure on resources and pollution governance, but also faces its constraints (Yuan et al., 2025). Environmental pollutants and CO₂ share common emission pathways in urban systems, generating correlated externalities in regional economy. Therefore, the collaborative treatment of carbon emissions and environmental pollution emerges as a necessary and effective path for sustainable development (Liu et al., 2025). Given the unidirectional transnational transfer of pollution-intensive industries, it is urgent for developing areas to explore the path of green development via trade cooperation across the global production-value network (Lee & Wen, 2025). Structural impediments—including overreliance on industry, asymmetrical access to green technologies in advanced and behindhand regions, and underdevelopment of factor markets in the global South cities—collectively impose systemic constraints on realizing balanced and harmonious green, low-carbon transition. This combination of global factors and structural characteristics makes the pursuit of the synergistic efficiency of pollution and carbon co-control (SEPC) a difficult and complex task for cities in developing countries.

Digital trade presents transformative potential to address these challenges. According to the UNCTAD Digital Economy Report 2024, the proliferation of digital infrastructure and technologies such as 5G, artificial intelligence, and blockchain has catalyzed **new data-based paradigms** in international trade and consumption. In fact, unprecedented growth in cross-border e-commerce, digital service trade, and data flows has become a new feature of today's digital economy. By dematerializing trade processes, facilitating cross-regional and cross-sector data/information flows (Li, 2023), and accelerating green technology diffusion, digital trade can strengthen regional innovation capacity and improve the efficiency of factor markets (Pomfret, 2020; Banker et al., 2011). Enhanced innovation supports green technology adoption and further structural adjustment (Wang et al., 2024), while a more efficient factor market reduces per-unit emission and asymmetry (Du et al., 2022). Digital trade therefore holds theoretical potentials for developing

regions to bypass green transition barriers. Clarifying and implementing these conceptual potentials remains critical. Establishing empirical clarity on the causality and affecting factors of digital trade and SEPC has become the key path for forming evidence-based strategies for UN Sustainable Development Goal 11 attainment.

Prevailing qualitative studies have widely acknowledged that digital trade has the potential to reshape trade structure, to enhance trade acting, and to broaden trade entities (Qi & Chu, 2022; Li et al., 2020). Moreover, by enhancing energy efficiency and amplifying scale effects, digital trade further advances the dual governance of pollution control and carbon reduction (Ashraf et al., 2024; Li et al., 2025; Jiang & Cai, 2024; Wilson et al., 2020). Most studies focus on digital trade at the national or provincial levels. When evaluating SEPC performance, it is a pity that existing studies does not pay enough attention to incorporate economic benefits comprehensively or neglect the coupled interactions among subsystems, resulting in an incomplete understanding of systemic development mechanisms. This study expands the analytical framework of digital economy by bridging trade mechanisms with local environmental governance via the perspectives of technology and market efficiency, and incorporating the dynamic view of rebound effect. By evaluating city-level digital trade and integrating economic benefits into local SEPC performance, this study provides nuanced empirical insights for systematically understandings of this topic.

This study investigates the green benefits of digital trade through a comprehensive analysis focusing on China—a pivotal member of the Global South—whose experiences offer valuable insights for emerging economies. First, this study evaluates digital trade and SEPC performance (integrating environmental and economic metrics) across 283 Chinese cities from 2013 to 2022. The direct impact is subsequently confirmed via both two-way fixed-effects (TWFE) and double-debiased machine learning (DDML) models. The results show that digital trade significantly enhances SEPC performance in China. To explore the impact mechanisms, we incorporate regional innovation capability and factor market distortions into the theoretical framework and calculate the precise mediation effect of each mechanism by combining DDML and structural equation modeling. This strategy reveals that regional innovation capability enhancement contributes 13.33% of the total effects, whereas factor market distortion alleviation contributes 24%. The threshold effect model demonstrates the existence of a nonlinear relationship between digital trade and SEPC.

The remainder of this study is organized as follows. Section 2 reviews the relevant literature on

this topic. Section 3 develops the theoretical framework and research hypotheses. Section 4 describes the research design, variables, and models. Section 5 presents the results and analysis of the empirical tests. The last section presents the conclusion and policy implications. **Figure 1** displays a detailed technical route of this work

Figure 1 *Technical route*

2.1 Definition, connotation, and realization paths of SEPC

environmental benefits as well as socioeconomic benefits and emphasizes the advanced nature of the concept compared with traditional environmental concepts. On this basis, some empirical studies have confirmed the existence of synergies between pollution and carbon reduction; for example, effective low-carbon policies can promote energy transition (Burandt et al., 2019), which is crucial for reducing air pollutants (Shindell & Smith, 2019). A further study compared the dual goals of pollution control with single-policy greenhouse gas reduction and reported that the dual goals of pollution and carbon reduction have greater health benefits and lower treatment costs (Zhang et al., 2021). This study highlights the synergistic function of the SEPC. Hence, the core path to achieving SEPC includes the synergistic effect of ecological and economic benefits, the coupling mechanism of pollution and carbon reduction, and cross-system synergistic effects.

Several paths are adoptable to achieve efficient SEPC. Reshaping energy, economic, and transportation structures is the first key point of pollution–carbon co-control (Lei et al., 2024). Moreover, the use of industrial parks has proven efficient in enhancing the synergistic effect of SEPC (Wang et al., 2024c). From the perspective of the digital economy, green technology innovation is recognized as a core driver of pollution–carbon synergistic governance (Liang et al., 2024; Zhai et al., 2024). Together, these studies reveal the central role of production and management innovations based on new digital technologies in reducing pollution and carbon emissions. The effective use of market mechanisms is also an important path for realizing SEPC. An evaluation of the carbon emissions trading system shows that market-based environmental regulation can significantly promote the synergistic effects of pollution and carbon reduction (Gan et al., 2024), and the development of green financial markets can help enhance green total factor productivity (Lee & Lee, 2022) and improve carbon emission efficiency (Wu et al., 2024) to help the synergistic governance of pollution and carbon reduction. At the industrial level, digital industrial agglomeration can significantly contribute to SEPC by promoting energy efficiency (Xu et al., 2024). The key pathways to achieving SEPC include **digitizing-based clean transformation** of production and management, efficient market mechanisms, and industrial upgrading and structure optimization. However, for developing economies, which are normally positioned as recipients of transnational industrial transfer and face a significant digital technology divide between the Global North and South, achieving SEPC requires extensive and effective international cooperation.

2.2 Green effects of digital trade

The unique nature of digital trade is harmonious with SEPC. Existing studies have emphasized the uniqueness of digital trade in terms of virtualization, platformization, and intensification (Tian et al., 2018). A newer review proposed four new perspectives for analyzing digital trade issues, namely, digital factor input, a new type of scale economy, a digital production network, and the digital platform economy (Chen et al., 2022). The key qualities of digital trade can be broadly categorized into two points: the digitization of trade methods and the digitization of trade objects. These two aspects jointly shape the potential of digital trade to promote SEPC by enhancing resource efficiency, reducing emissions in the process, and enabling cleaner production through digital technologies.

Digital trade nourishes an environment conducive to green growth for developing economies. Macrolevel studies have revealed that digital trade can help overcome the middle-income trap by redirecting market investment, promoting consumption upgrading, and fostering new export advantages (Qi & Chu, 2022). In cross-border trade, increased digital intensity can significantly reduce trade costs (Chiappini & Gaglio, 2024), increase the amount of technological value added and likely change the value distribution of traditional value chain curve theory (Li et al., 2020), which helps realize the upgrading of the global value chain position (Wu et al., 2024). In intratrade, digital trade is able to overcome the limitations of administrative and geographic boundaries and help breakdown regional trade barriers and market segmentation (Li, 2023).

By reshaping industrial ecosystems, digital trade facilitates the clean transition of production and management. Studies at the meso-level have noted that the digitization of trade entities can help increase the complexity of exports (Atasoy, 2021). Additionally, digital trade reshaped the pattern of industrial agglomeration. A study on the creative industry reveals the characteristics of virtual agglomeration, which can break through the spatial and temporal constraints of traditional geographic agglomeration, mitigate monopoly power, enhance competition in the industry, and simultaneously form valuable data resources and promote the in-depth integration of industries.

Digital trade empowers core actors within the green ecosystem. The literature from the micro perspective reveals that digital trade helps technological innovation via the spillover effect (Zhang et al., 2023), helps optimize resource allocation to increase the operational efficiency of firms and promotes the integration of standardization and ICT to increase the efficiency of

firms' supply chains (Saengchai & Jermstittiparsert, 2019). It also empowers the efficient allocation of trade credit (Zhou & Li, 2023).

2.3 Interactions between digital trade and SEPC

Existing studies show that digital trade can significantly suppress carbon emissions by upgrading the consumption structure of rural residents (Wang et al., 2023) and effectively reduce the long-term ecological footprint, contributing to the maintenance and restoration of the ecological carrying capacity (Ashraf et al., 2024). In addition, digital trade can contribute to SEPC in the early stage by improving energy efficiency, realizing scale effects, promoting technological innovation and structural optimization and upgrading (Li et al., 2025). As an important subcategory of digital trade, the digital service trade involves similar carbon-reducing mechanisms (Jiang & Cai, 2024). From a broader perspective, digital trade contributes to the diffusion of digital technologies, which promote the greening of exports, leading to the development of low-carbon trade (Wang et al., 2024b; Wen & Zhu, 2024) and more green commodities on the consumer side, with potential environmental benefits (Wilson et al., 2020). Hence, digital trade plays a positive role in SEPC through various channels, such as improving energy utilization efficiency, promoting technological innovation, and facilitating information flow.

However, the recently emerging concept of the “digital rebound effect” provides a more nuanced view to the interaction between digitalization and the environment. The most intuitive logic is that increasing electricity consumption caused by large scale digital infrastructure and services may offset the energy-saving and emission-reduction effects of digitalization, leading to an increase in final energy consumption and carbon emissions (Liu et al., 2025; Zhu & Lan, 2023). Like other effects of digitalization, the rebound effect is also influenced by factors such as urbanization and industrial structure (Huang & Lin, 2025), and it also exhibits spatial spillover effects (Qian et al., 2025). Academic debates in this area mainly focus on the timing of the emergence of the “backfire” effect. Some literature suggests that there is a U curve between digitalization and energy consumption, so as the associated environmental degradation, meaning that an appropriate scale of digitalization in the early stage is conducive to energy conservation and emission reduction (Lan & Zhu, 2023); another branch of literature argues that the “backfire” effect even appears in the early stage of digitalization and gradually diminishes as digitalization deepens (Peng et al., 2023).

It can be seen that the interaction between digitalization and environmental governance is still a

tricky topic. More in-depth research is needed to supplement the estimation of the total effect and structural comparison.

2.4 Literature synthesis and research gaps

The literature on digital trade and SEPC highlights the role of digital trade in green development. It fosters technological innovation, facilitates technology spillovers, and supports green technology advancement for pollution and carbon reduction. Additionally, digital trade promotes digital technology adoption, enhances market efficiency, and optimizes resource allocation for SEPC. Industrial digitalization and virtual aggregation further promote industrial upgrading and strengthen pollution-carbon governance.

The literature provides many theoretical clues about the promoting effect of digital trade on SEPC, indicating the feasibility of the study in this paper. However, existing studies seldom measure digital trade at the municipal level and often overlook economic benefits in SEPC evaluation. Research directly linking digital trade to SEPC remains limited. Hence, this study draws on the implication of the literature, remeasures the development of digital trade at the prefecture level, considers economic benefits as one of the subsystems to be included in the measurement of SEPC, and directly investigates the impact of digital trade on SEPC to provide more theoretical insights into the causes of digital trade and environmental protection.

3. Theoretical analysis and hypothesis proposal

The core theory of digital trade-environment nexus in this study is mainly an extension and reconstruction of classical international trade theory in the digital context, while retaining the environment economics tradition of integrating multi-agent, multi-level, and multi-disciplinary theories. From the perspective of trade cost and industrial organization theories, the unique platform-network structure of digital trade reduces transaction costs, reshapes market organization, and improves production efficiency. Drawing on endogenous growth and technology diffusion theories, digital trade facilitates knowledge spillovers and competition-driven innovation, thereby enabling productivity-enhancing and cleaner production. Moreover, market failure theory suggests that digital trade mitigates factor market distortions by reducing information asymmetry and mobility barriers, supporting green structural transformation. However, efficiency gains from digitalization may induce rebound effects that partially offset environmental benefits, implying potential scale constraints on digital trade. **Figure 2** step 3 provides an illustration of this

multidimensional theoretical texture, the following subsections provide detailed analyses.

3.1 Platform-network structure of digital trade and green efficiency

Consistent with trade cost theory, the digitalization of trade objects fundamentally weakens the traditional linkage between economic expansion and material resource consumption. New digital products, including data, content, and technical services, do not require large consumption of material resources and therefore avoid waste and pollution emissions in the traditional industrial production process (Wilson et al., 2020). Moreover, digital production relaxes the spatial concentration emphasized in new economic geography models, and the decentralized remote online production mode can reduce road transportation and energy consumption, thus reducing carbon emissions (Chiappini & Gaglio, 2024). From the perspective of trade mode digitization, international trade has a complex network structure, and import and export activities are mainly matched through the existing trade network (Chaney, 2014). Within a network-based trade framework, digitized trade modes reduce matching frictions and trade costs while expanding effective market access, thus realizing the co-optimization of resource savings and economic efficiency.

From the perspective of industrial organization theory, the platform-based structure of digital trade reshapes market organization to facilitate SEPC. The platform economy of digital trade can promote the integration and complementarity of economies of scale and economies of scope; strengthen the coupling of the same industry, related industries and related industrial clusters; further expand the resource sharing and industrial collaboration within the clusters; and enhance the efficiency of the regional economy (Xu et al., 2024). Moreover, consistent with agglomeration externality theory, the application of digital information technology can enhance the diffusion of information and the pooling efficiency of the labor force within clusters (Saengchai & Jermittiparsert, 2019), expand the positive externalities of industrial agglomeration, and ultimately increase the productivity of regional industries. This means that digital trade can enable the regional economy to achieve higher output on the basis of the same energy consumption, thus realizing the synergistic development of pollution reduction, carbon reduction and economic efficiency.

At the enterprise level, digital trade operates as a channel of endogenous technological progress. The learning by doing effect involves the process of carrying out digital trade accompanied by the widespread use of digital technology at the enterprise level (Zhang et al., 2023). Once enterprises

embed digital technology into their production and management processes, they can utilize data elements to connect the “production silo”, thus realizing unified scheduling and coordinated matching of production resources and thus achieving the simultaneous improvement of cleaner production and production efficiency (Chong et al., 2024). Accordingly, endogenous productivity improvement brought by digital trade can directly enhance SEPC through emission intensity reduction rather than scale expansion alone. Hence, the following hypothesis is proposed.

Hypothesis 1. Ceteris paribus, digital trade can directly promote regional SEPC.

3.2 Technology diffusion and regional innovation capability

The innovation-mediated effect of digital trade on SEPC is primarily grounded in trade-based technology diffusion theory, which emphasizes the role of international trade as a carrier of knowledge diffusion across regions, industries, and firms. Digital trade includes digital products and services on the basis of traditional trade and widens the range of trade objects, thus expanding the effect of technology transfer in the trading process (Wen et al., 2023). For international diffusion, digital trade increase the availability of global innovation resources, further weaken the spatial and temporal barriers to cross-border trade, accelerate technological spillover, and help enhance regional innovation capacity (Li, 2023). For industrial diffusion, digital trade platforms promote the diversified integration of “digital + services + manufacturing” (Tian et al., 2018), facilitates the cross-industry integration of heterogenous knowledge, and plays a positive role in interindustry technology transfer. For demand-supply diffusion, digital trade directly links the supply and demand sides, which can more effectively convey demand information to the supply side, promote product renewal and iteration, and strengthen the motivation for the R&D of regional innovation subjects.

Regional innovation is a key condition for SEPC. From the supply and demand sides, innovation can intuitively provide green production methods for industry and agriculture, replacing the high pollution and high emissions of the traditional production process (Finger, 2023). New technologies provide green products for green consumption and green life (Tariq et al., 2017), creating new economic growth points, thus balancing environmental governance and economic development and realizing the synergistic effect of the SEPC. Digital innovation is an important channel for carbon neutrality, and it reduces energy consumption under the same output conditions (Zhou et al., 2024). Relying on new digital techniques, the environment governance network is now a feasible solution

for realizing the synergistic control of pollutants and carbon emissions (Feroz et al., 2021); this network also reduces the costs of environmental regulation and pollutant disposal, achieves higher efficiency with lower costs in green development (Maiurova et al., 2022), and then impacts the SEPC. **Through trade-induced technology diffusion, digital trade ultimately enhances SEPC.** Hence, another hypothesis is proposed.

Hypothesis 2. Digital trade promotes SEPC by fostering regional innovation capabilities.

3.3 Market failure correction and green productivity

The market failure theory emphasizes information asymmetry, imperfect competition, and institutional barriers as key sources of inefficient factor allocation. **First, digital trade alleviates information asymmetry.** Digital trade, including digital service trade, improves interest integration and transaction efficiency of the global production network (Pomfret, 2020) and accelerates the wide-area agglomeration and transformation of factors. Digitized trade methods feature a flattened structure and fewer intermediate links, which can improve information asymmetry and consumer trust (Banker et al., 2011), thus easing obstacles to the free flow of capital. **Second, digital trade mitigates institutional and structural barriers in labor markets.** Digital trade creates new types of jobs, increases total employment, and activates the labor factor market; it also cultivates new types of industries, promotes industrial structural adjustment, prompts the flow of labor between industries, and achieves better matched marginal output (Qi & Chu, 2022) to ease labor factor distortion. The digitization of labor content and labor modes can effectively overcome the barriers to human resource mobility, such as household registration systems and geographic barriers (Li, 2023), and alleviate the distortion of labor factors. **Third, digital trade weakens imperfect competition.** Industrial digitization and digital industrialization can change the conditions of certain natural monopoly industries and strengthen market competition, thus accelerating the free flow and optimizing combinations of production factors.

By correcting market failures in factor allocation, digital trade improves the marginal productivity of labor and capital and enhances total factor productivity, which helps balance SEPC and economic development. Improved factor allocation and newly added data factors can promote the upgrading of the global value chain (Zou & Lei, 2023), thus increasing the added value of products and coordinating the reduction of carbon emissions and economic development. Factor distortion hinders industrial structure upgrading and negatively moderates the positive effect of green

technology innovation on industrial structure (Wang et al., 2024), so the improvement of factor distortion can promote industrial upgrading, thus changing the traditional industrial structure of high energy consumption and high pollution and subsequently promoting the SEPC. In summary, digital trade can ease factor distortions and thus positively contribute to the development of SEPC. Therefore, a third theoretical hypothesis is proposed.

Hypothesis 3. Digital trade promotes SEPC by mitigating factor market distortions.

3.4 Rebound effects and the optimal scale of digital trade

The digital rebound effect refers to the phenomenon whereby advancing digitalization may induce elevated energy consumption and concomitant emissions, caused by the development of energy-intensive digital infrastructure and digital industries (Liu et al., 2025). This theoretical perspective posits that an optimal scale of digitalization exists to strike a balance between new economic growth and environmental efficiency (Lan & Zhu, 2023). As a specific application of digitalization in economic sectors, digital trade is likely to obey the same logic. It is clear that digital trade evolves through two asynchronous stages: namely, the digital transformation of conventional trade industries and trade forms, and the emergence of trade activities within emerging digital sectors. The former stage focus on efficiency improvement, whereas the latter stage relies heavily on energy-intensive infrastructure including data centers and computing facilities to create trade growth. Collectively, the digital rebound dynamics imply that the positive impact of digital trade on SEPC may diminish to a certain content as it expands to a larger scale.

Hypothesis 4. Digital trade exhibits an inverted U-shaped impact on SEPC due to the digital rebound effect.

4. Research design

To conduct scientific and comprehensive examination of the proposed hypotheses requires a multi-step workflow. **Figure 2** illustrates the organization and main content of each step of this study.

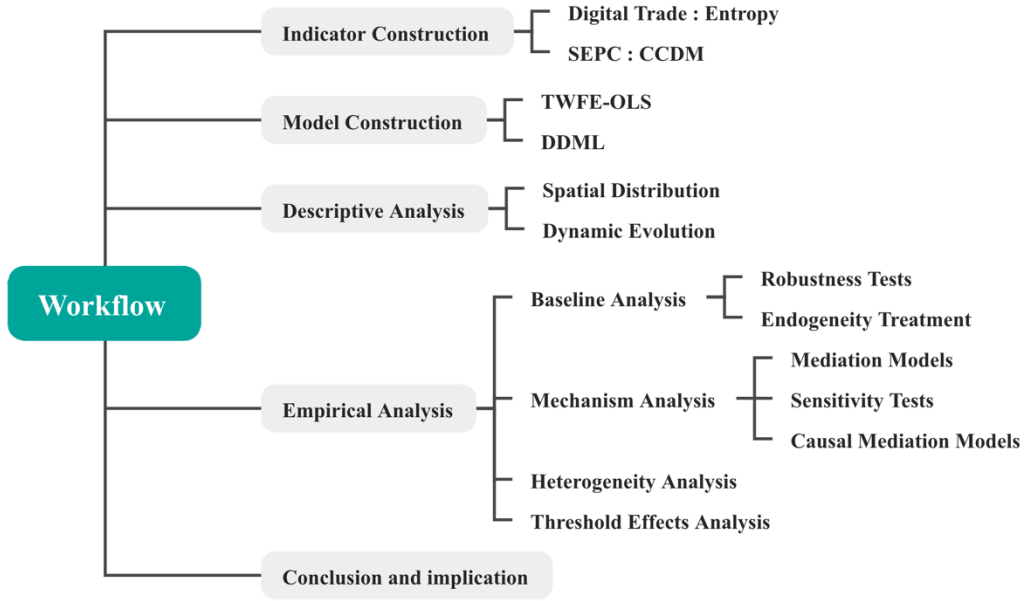


Figure 2 Workflow of empirical analysis

4.1 Empirical model

Considering that there are significant individual characteristics of each city and each year, it is necessary to fix the individual effect and time effect to obtain a more pure and reliable estimation. The model is set as follows on the basis of the research hypotheses:

$$CCD_{it} = \alpha_0 + \beta DTI_{it} + \sum_{k=1}^K \theta_k X_{k,it} + \gamma_t + \delta_i + \varepsilon_{it} \quad (1)$$

where CCD_{it} and DTI_{it} denote the degree of coupling between the SEPC and digital trade development of city i in year t , respectively. β is the coefficient of interest in this paper. γ_t and δ_i denote the year fixed effects and city fixed effects, respectively, representing unobservable characteristics specific to years and cities. ε_{it} represents the random disturbance term.

The control variables (X_{it}) include the proportion of secondary and tertiary industry (Sec_Ind , Trd_Ind), the number of industrial enterprises above scale (ln_Ind), the number of foreign-invested enterprises (ln_Fdi), science and technology fiscal expenditures (ln_Sci), and the balance of financial institutions' loans (ln_Fin). The proportion of secondary industry reflects cities' economic development mode, and the proportion of tertiary industry reflects advanced industrialization, which has a far-reaching impact on both environmental control and economic growth. As the main source of pollutants and greenhouse gases, the number of industrial enterprises plays a direct role in SEPC; at the same time, industrial enterprises are also the main source of taxes for cities, so this variable

affects regional economic development. According to the pollution paradise hypothesis, FDI, to a certain extent, reflects potential pollution diversion. Science and technology fiscal expenditures reflect local governments' willingness and ability to intervene in high-quality development, which will play a role in the development of SEPC. The balance of financial institutions' loans reflects the financial activities of the city, which affect the pollution emission intensity and economic vitality.

As a nonparametric estimation method, DDML does not preset the functional form of the model, so it can better capture the nonlinear effect than the traditional linear models. DDML also avoids the “Dimensional Curse” caused by control variables via residual regression estimation, which effectively compensates for the shortcomings of traditional causal inference. In this paper, we construct a supervised DDML model as follows. Traditional TWFE models are used to cooperate with DDML to obtain a reliable estimation of the treatment effects.

A partial linear model was first set up as Equation (2) to release the linearity assumption of the control variables. where function g is the form of the control variables affecting DTI . $\hat{g}$ is estimated via a machine learning algorithm to obtain $E[CCD_{it} | X_{it}]$. At this point, through the auxiliary model in Equation (3), we can calculate $\hat{f}$ and then obtain $E[DTI_{it} | X_{it}]$. Therefore, the coefficient of interest θ_0 can be estimated by the residual regression model in Equation (4), which is free from possible bias caused by control variables.

$$CCD_{it} = \theta_0 DTI_{it} + g(X_{it}) + \varepsilon_{it}, E[\varepsilon_{it} | X_{it}, DTI_{it}] = 0 \quad (2)$$

$$DTI_{it} = f(X_{it}) + \mu_{it}, E[\mu_{it} | X_{it}] = 0 \quad (3)$$

$$(CCD_{it} - E[CCD_{it} | X_{it}]) = \theta_0 (DTI_{it} - E[DTI_{it} | X_{it}]) + \sigma \quad (4)$$

DDML introduces a K-fold cross-fitting method to address overfitting, which equates the data into K subsets, uses one of the (K-1) subsets to train the function, calculates the residuals via the remaining subset, and repeats the step to avoid overfitting caused by data noise. Since random sampling is involved, this paper sets the random seed as 123, and the number of folds K of the sample splits are all 5, unless otherwise specified.

4.2 Variable definitions

4.2.1 Digital trade index (DTI)

Quantifying digital trade development is a complex task, which involves economic interpretability, data availability and index representativity. Due to very limited data availability, existing researches mainly focus on cross-national level. The straightest index is the amount of

digital trade and digital service trade provided by the OECD, this dataset only covers 74 countries from 2014 to 2020 (Suh & Roh, 2023), which is insufficient to support mainstream micro researches. Other available data are basically national-level e-commerce sales, trade volumes of digitally delivered services, etc., and there are no very intuitive data available. For example, in Ashraf's (2024) study, the percentage of total trade in digital deliverables and services was used as a proxy variable for digital trade. Another example is Guo et al. (2024), who used the exports of ICT enabled services as a proxy variable for digital trade. Yet another example is Wang et al.(2025), who subjectively defined the 'digital trade' industry and then produced a industry-specific trade dataset for their research. It's clear that the several relatively intuitive trade data cannot support in-depth research of cities' digital trade.

To quantify digital trade development with an index system is a common practice in academia. Also there are significant commonalities in these published index systems. After review, we believe that these commonalities are able to represent some basic concepts that must be covered when measuring digital trade, such as digital infrastructure, the environment to conduct trade, the digitalization of existing industrial trade, and the trade of new digital industries (Wang, 2023; Wu et al., 2024; Zhu et al., 2025). Adhering to the principle of prudent science, the indicator system we have established is basically the intersection of these systems and should be able to reflect the level of urban digital trade.

In **Table 1**, per capita GDP, social consumption expenditures on retail sales, road freight volume, and foreign trade dependence provide a multifaceted reflection of the general commercial development of a city (Li et al., 2020). These indices serve as fertile ground for digital trade development, whereas the mobile phone penetration rate, internet accessibility, and proportion of employment in information software-related industries represent a region's ability to adopt digitization of trade modes, production of digital trade targets, and overall potential for regional digital trade development (Zhou et al., 2022). The e-commerce sector represents a tangible manifestation of industrial digitization, as it transforms traditional trade into its digital form (Gao et al., 2025). Software business and IT services are indicative of industry digitization, serving as indicators of a region's capacity to develop digital trade within its boundaries (Liang et al., 2024). These indicators provide a comprehensive and in-depth evaluation of regional digital trade development levels.

This paper adopts the entropy weighting method to objectively synthesize the index system. The information entropy weight of each indicator is given in **Table 1**.

Owing to subindex and entropy weighting limitations, several shortcomings of this index system must be discussed. First, entropy weighting may not reliably reflect variables' true relative importance across all scenarios, it assigns weights based on data balance, risking lower weights for more balanced variables that hold more valuable economic information. Second, critical variables may be omitted. As adding variables introduces correlated or independent measurement errors and escalates complexity, meaningful variables like cross-border data flows and digital platforms could be omitted, leading to misestimation of digital trade's actual coefficients. Yet this issue rarely undermines conclusions when the research is limited on causal direction rather than precise prediction. Third, economic interpretability is not guaranteed. Aggregating multiple indicators may turn the system into a 'black box', eliminating interpretability of correlations in specific domains. Notably, these limitations do not invalidate the study's causal inference, they merely delineate the theoretical boundaries of this work and point to avenues for future index system optimization.

Table 1 Indicator system and weights for calculating DTI

Indicators	Sub-indicators	Unit of measure	Entropy weights
Digital trade environment	GDP per capita	Billion yuan	0.034
	Total retail sales of consumer goods	Billion yuan	0.068
	Highway freight volume	Billion tons	0.039
	Foreign trade dependence	%	0.059
Digital trade potential	Mobile phone users per 100 people	People	0.033
	International internet users per 100 people	People	0.035
	Urban employment share in information transmission, software, and IT services	%	0.035
	Digital inclusive finance index	-	0.019
Industrial digitalization	Proportion of enterprises engaged in e-commerce transactions	%	0.075
	E-commerce sales revenue	Billion yuan	0.090
	E-commerce procurement revenue	Billion yuan	0.092
Digital industrialization	Software business revenue	Billion yuan	0.128
	Software product revenue	Billion yuan	0.118
	IT service revenue	Billion yuan	0.129
	Telecommunication business revenue per capita	Billion yuan	0.047

4.2.2 Coupled coordination degree of SEPC (CCD)

SEPC extends traditional pollution control by emphasizing system-wide interactions between industrial pollution and carbon emissions. It highlights the need to couple these factors while balancing environmental protection with economic growth. Since pollution reduction is proposed by public demand for a better quality of life, SEPC seeks a harmonious approach that integrates pollution control, carbon management, and economic development.

This study adopted a CCDM to evaluate the synergistic effect of SEPC. The coupled coordination degree is composed of two components: the system development level (T) and the system coupling degree (C). The final value represents the geometric mean of T and C, providing a comprehensive representation of the overall system's developmental stage and the degree of coordination among its subsystems. In Equation (5), to mitigate the risk of the T value being excessively large and prioritize the coordination of ecological and economic components, this paper sets $\theta_1 = \theta_2 = \theta_3 = \theta_4 = \theta_5 = 0.2$ in strict accordance with the practice in prevailing mainstream literature (Fan et al., 2019; Han et al., 2023; Wang et al., 2019; Xing et al., 2019; Yang et al., 2020; Yin & Xu, 2022). Specifically, equal weights implicitly assumes that each subsystem plays an equally important role in the process of synergistic evolution, which is critical to focus the indicator on balance and coordination. Au contraire, differentiated weighting could obscure the clarity of coupling coordination. Also, in the absence of strong a-priori evidence supporting differentiated weighting, assigning equal weights helps to minimize potential subjective bias. In Equation (6), $Subsystem_{it}^n$ represents the same five components as in Equation (5).

$$T_{it} = \theta_1 GDP_{it} + \theta_2 SO2_{it} + \theta_3 Dust_{it} + \theta_4 Effluent_{it} + \theta_5 CO2_{it} \quad (5)$$

$$C_{it} = \left(\prod_{n=1}^5 Subsystem_{it}^n \right)^{\frac{1}{5}} / \frac{1}{5} \sum_{n=1}^5 Subsystem_{it}^n \quad (6)$$

$$CCD_{it} = \sqrt{T_{it} \times C_{it}} \quad (7)$$

4.3 Sample selection and data sources

This study selected relevant data from each prefecture-level city across China, carefully removed massive missing values, and used interpolation to fill in the gaps for variables that were only partially missing. After careful processing, all the variable means, maximums, and standard deviations showed no significant changes. The final panel dataset included 283 city-level datasets spanning from 2013 to 2022, sourced from official statistics yearbooks published by the National Bureau of Statistics of China, as well as reputable databases such as CSMAR (China Stock Market & Accounting Research Database) and CNRDS (Chinese Research Data Services). Detailed explanations for the variables used are provided in **Table 2**.

Table 2 *Variable definitions*

Types	Symbols	Variables	Definitions
Dep. Var.	<i>CCD</i>	SEPC	A standardized index calculated using CCDM
Ind. Var.	<i>DTI</i>	Digital trade	A standardized index obtained using entropy weighting
Control Variables	<i>Sec_Ind</i>	Proportion of second industry	Second industry added value/regional GDP
	<i>Trd_Ind</i>	Proportion of tertiary industry	Tertiary industry added value/regional GDP
	<i>ln_Ind</i>	Potential pollution intensity	Logarithm of the number of industrial enterprises above scale
	<i>ln_Fdi</i>	Pollution diverse intensity	Logarithm of the number of foreign-invested enterprises
	<i>ln_Sci</i>	Government intervention	Logarithm of fiscal expenditure on science and technology
	<i>ln_Fin</i>	Economic vitality	Logarithm of the total balance of loans from financial institutions
Mediating Variables	<i>Invention</i>	Regional innovation capability	The number of invention patent applications
	<i>Utility</i>	Factor market distortion	The number of novel utility patent applications
	<i>FMD</i>		The degree of factor market distortion is calculated based on the Cobb–Douglas production function

4.4 Descriptive statistics

Owing to the differences in calculation methods for *CCD* and *DTI*, as well as unit variations in secondary indicators, there was a significant disparity between the two core indicators. Moreover, the values of composite indicators lack direct economic meaning and comparability. To eliminate dimensional disparities and enable coefficient comparison, this study standardized *CCD* and *DTI* through normalization processing. For the control variables, notable discrepancies exist among continuous variables such as government expenditures, loan balance, and enterprise numbers, so logarithmic analysis was employed to address these issues. To mitigate the impact of extreme values on regression estimates, all variables underwent 1% trimming above and below. Descriptive statistics for the treated data are presented in **Table 3**. The processed data retained consistency at the observation scale, with mean values within reasonable ranges and minimal intervariable disparities, standard deviations conforming to typical characteristics, and extreme values falling within normal intervals, which is consistent with economic logic.

Table 3 *Descriptive statistics of the variables*

Variables	Obs	Mean	Sd.	Min	Max
<i>CCD</i>	2733	0.689	0.091	0.000	1.000
<i>DTI</i>	2733	0.139	0.184	0.000	1.000
<i>Sec_Ind</i>	2733	43.525	10.540	10.680	79.360
<i>Trd_Ind</i>	2733	44.380	9.542	16.440	83.870
<i>ln_Ind</i>	2733	6.681	1.069	3.045	9.536
<i>ln_Fdi</i>	2733	2.967	1.636	0.000	8.099
<i>ln_Sci</i>	2733	1.434	1.452	-2.586	6.319
<i>ln_Fin</i>	2733	7.545	1.152	4.741	11.477
<i>did</i>	2733	0.112	0.315	0.000	1.000
<i>Invention</i>	2733	0.025	0.069	0.000	1.000
<i>Utility</i>	2733	0.036	0.081	0.000	1.000
<i>FMD</i>	2733	0.463	0.038	0.001	0.520

Figure 2 visualizes the spatiotemporal evolution of digital trade levels in Chinese cities. Taking 2013 as the baseline year, it is evident that the proportion of cities in the upper quartile (Q4, 75%-100%) increased rapidly. By 2022, 59.1% of sample cities had reached a level equivalent to Q4 2013. However, the rapid growth of digital trade across Chinese cities has been uneven. The ridge line plot in the lower-left corner indicates a gradual decline in the peaks of kernel density curves over time, revealing widening inter-city disparities. Spatially, the maps demonstrate that digital trade advancements have been primarily concentrated in coastal cities and industrial hubs in the eastern and central regions, while Northeast China and western regions with less developed industrial bases have made limited progress. The standard deviation displays an accelerating upward trajectory, indicating potential “Matthew effects” in digital development. While both the mean and median of *DTI* demonstrate steady increase over time, this growth appears increasingly uneven.

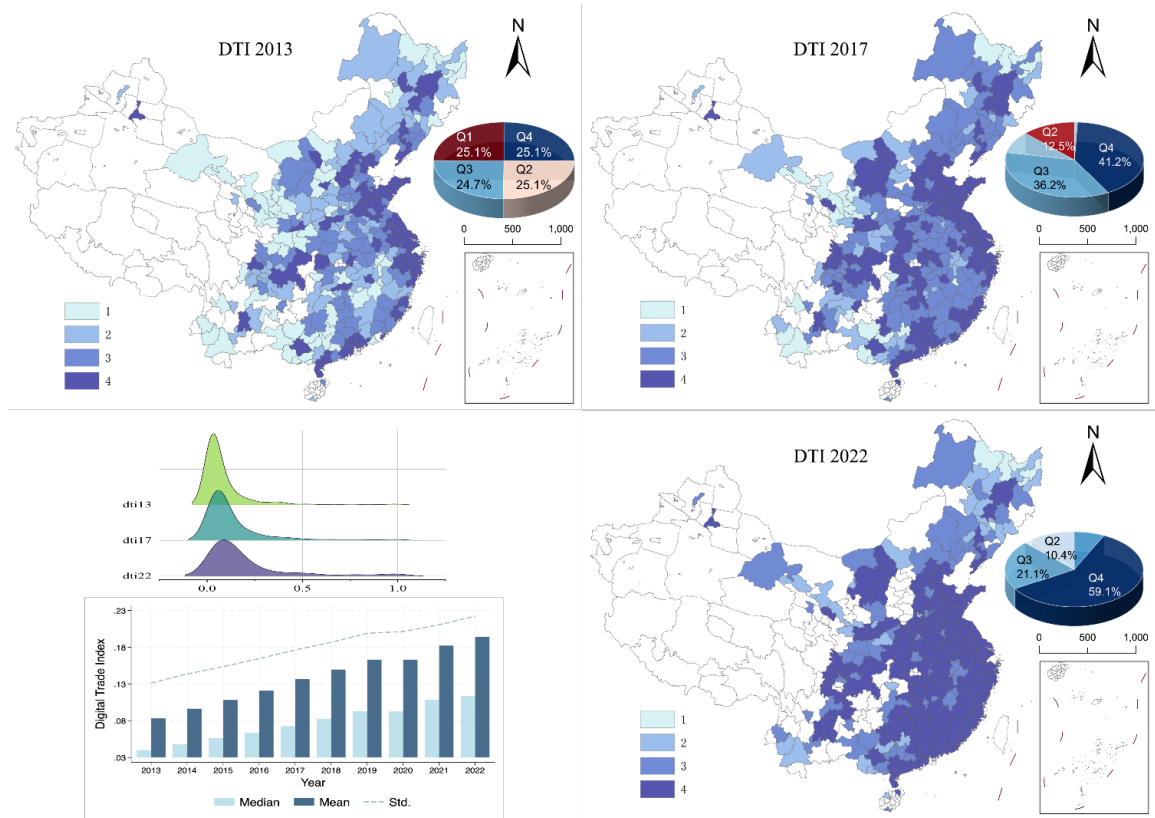


Figure 3 Visualization of *DTI* evolution

Figure 4 provides a clearer visualization of the spatial imbalance in *DTI* distribution. Initially, the 2017 map reveals a relatively balanced distribution across all four quartiles, with several dozen cities clustered within the 75%-100% range. By 2022, however, the vast majority of cities concentrated in the last quartile, while only a minimal number achieved higher *DTI* values. Combined with **Figure 3**, these panels demonstrate a distinctively centralized, hub-oriented

development pattern in China's digital trade landscape. The radar plots detail the spatial evolution: **eastern and central regions contributed most economic growth**, whereas Northeast China even experienced a decline, confirming our earlier hypothesis regarding regional divergence. The second spider chart compares *DTI* trajectories across ten major city clusters in China. Notably, the Pearl River Delta cluster outperforms the Yangtze River Delta cluster in terms of digital trade development despite both being economic powerhouses. While the Chengdu-Chongqing (CY) and West Taiwan Strait (WTS) clusters maintain stable growth, the Shandong Peninsula (SD) cluster experiences volatility, and the Central-Southern Liaoning (CSL) and Beijing-Tianjin-Hebei (JJJ) clusters show *DTI* regression. This observation, to some extent, reflects the profound influence of port cities' trade advantages and traditional commercial foundations on shaping the evolution of contemporary digital trade.

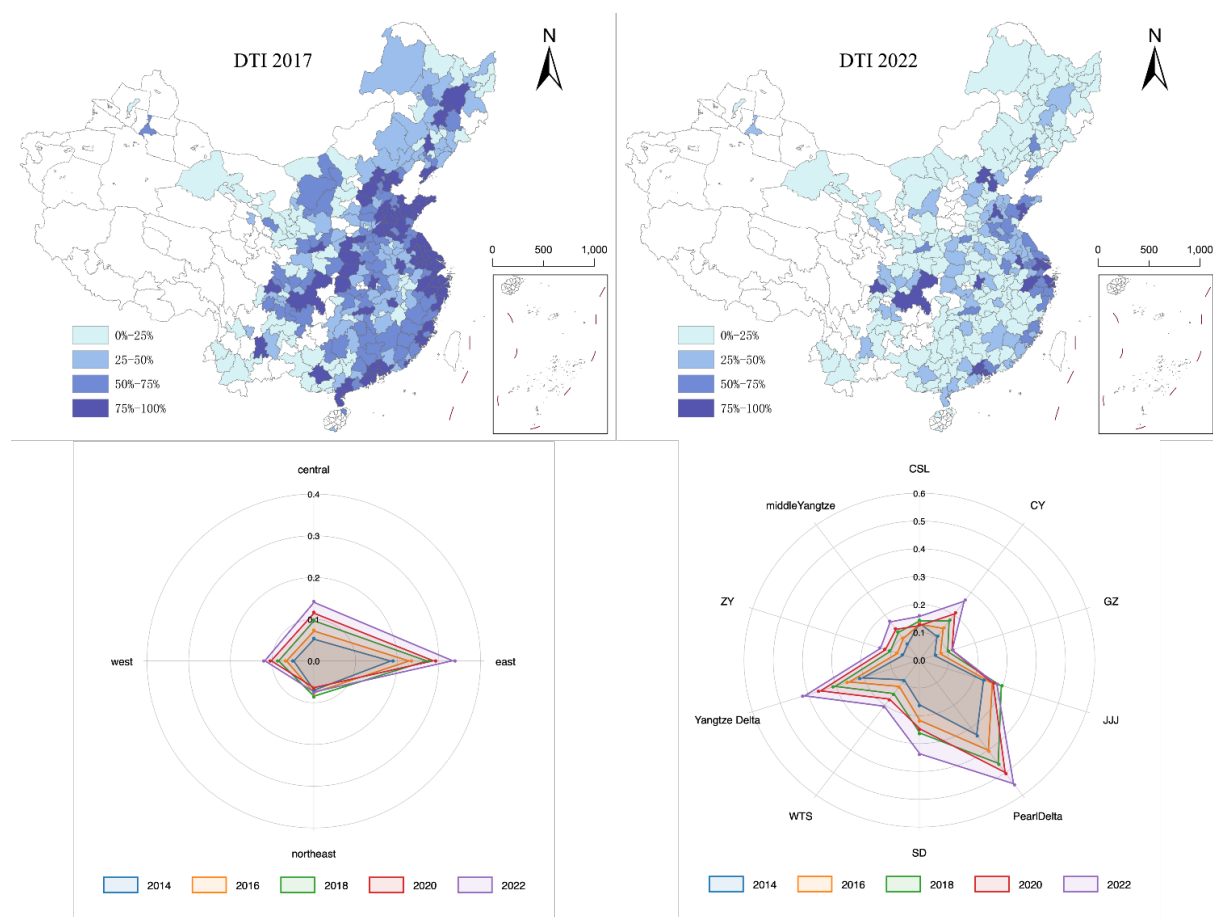


Figure 4 *DTI development of city clusters*

Figure 5 demonstrates substantial improvement in Chinese cities' SEPC, with 55.9% of sample cities having transitioned into the level equivalent to Q4 2013 by 2022. Notably, the peaks of CCD's kernel density curves remain stable while its standard deviation curve exhibits only gentle slope,

suggesting relatively balanced development of SEPC across Chinese cities and potentially indicating the absence of pronounced spatial growth poles.

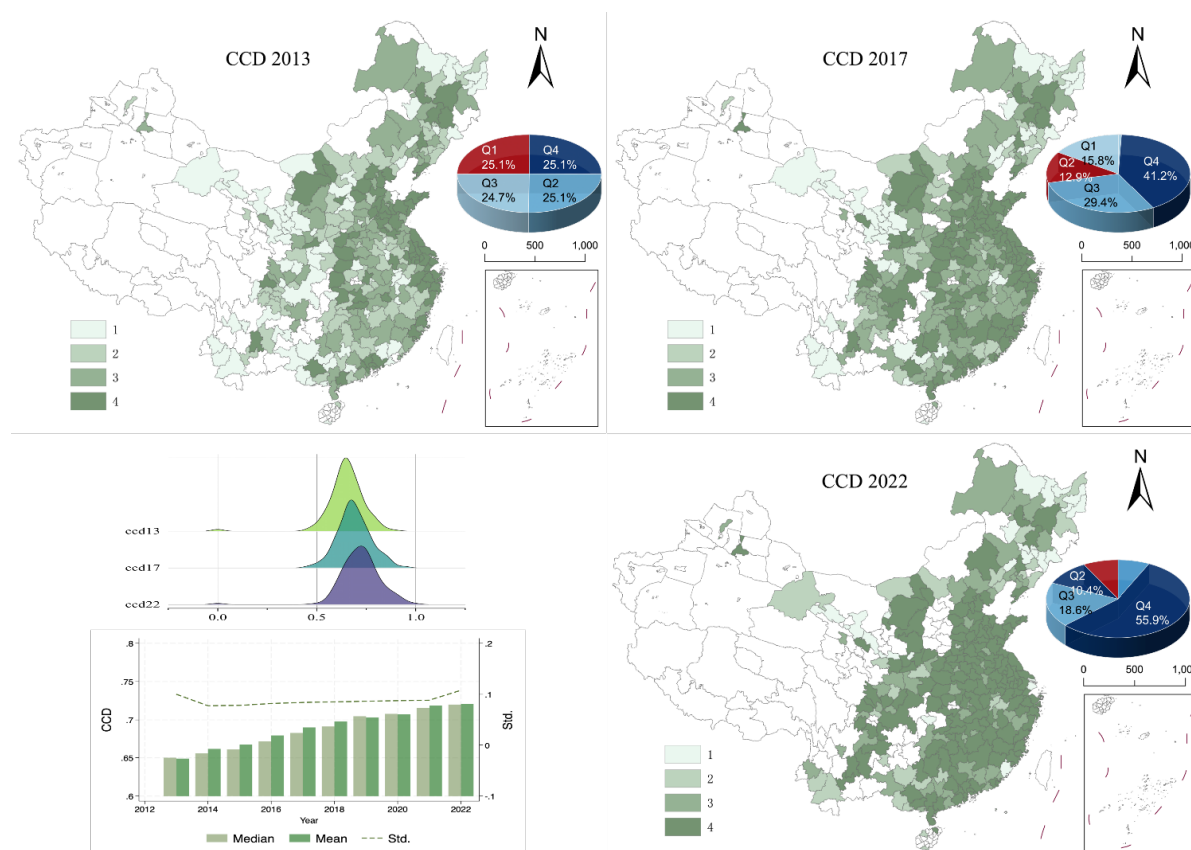


Figure 5 Visualization of the SEPC evolution

Figure 6 reveals consistent distribution of cities across the four quartiles in both maps, indicating relative stability in Chinese cities' SEPC development trajectory. The scarcity of inter-city exceeding demonstrates the foundational role of constant natural ecological endowments in SEPC development. The radar charts in **Figure 5** offer detailed spatial analyses. Notably, the performance hierarchy of ten major clusters in SEPC diverges from their *DTI* rankings yet reveals strategic interconnections. For instance, the JJJ cluster exhibits SEPC improvements despite *DTI* regressions, likely reflecting green transition strategies in Beijing and Tianjin. Conversely, the Yangtze River Delta cluster excels in *DTI* but exhibits stagnant SEPC growth, mirroring the subtle declines observed in the WTS cluster. These cases thus reveal existing trade-offs between SEPC and digital economic development in Chinese cities. This necessitates urgent exploration of innovative mechanisms to harmonize ecological construction with digitalization.

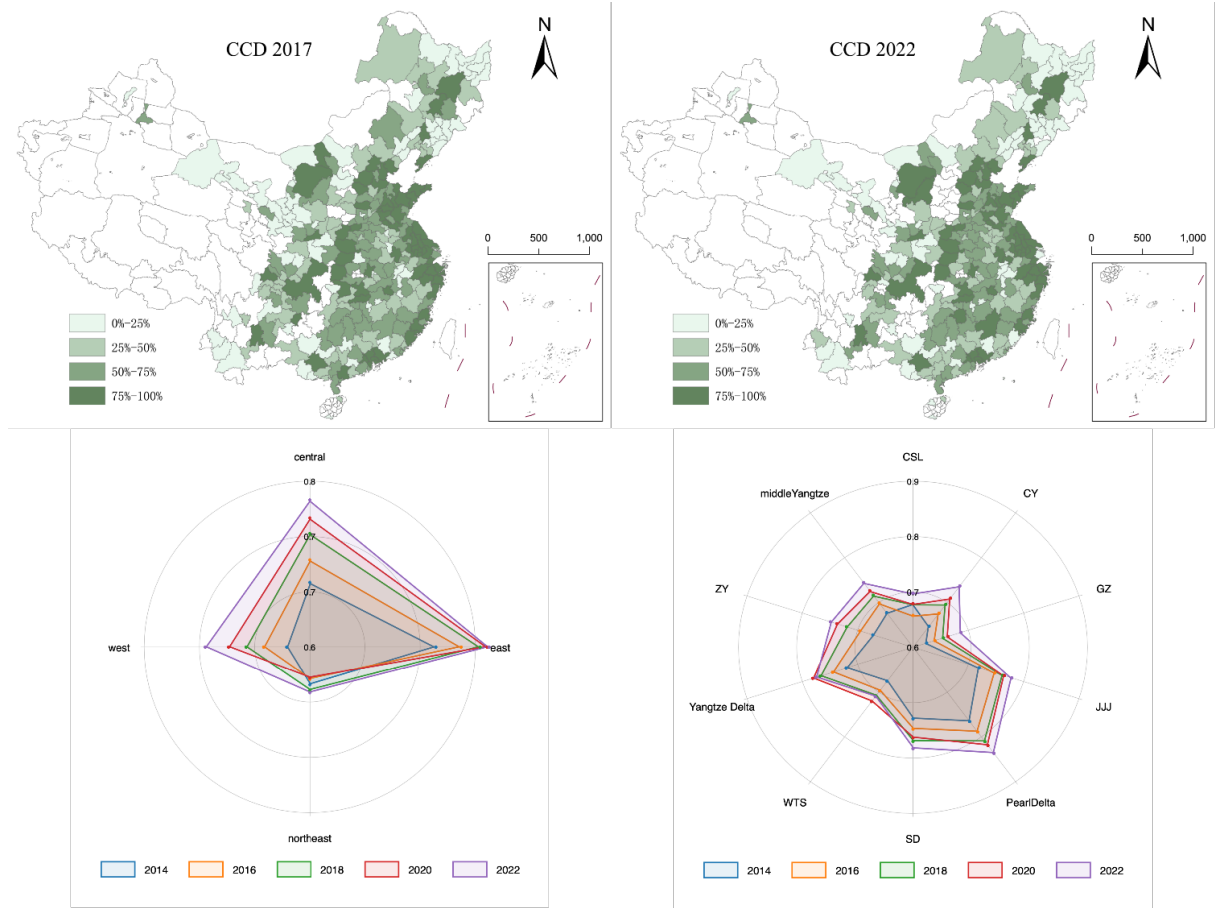


Figure 6 SEPC development of city clusters

Figure 7 visualizes the bivariate distributions of *DTI* and *CCD* across all samples, with green projections on the left surface indicating their correlation. The upward-sloping scatter plot configuration confirms a positive *CCD-DTI* association, and the slope of the univariate pooled linear regression is 0.349, which aligns with prior theoretical expectations. However, rigorous econometric examinations and analyses remain essential to establish causal linkages beyond this observed correlation.

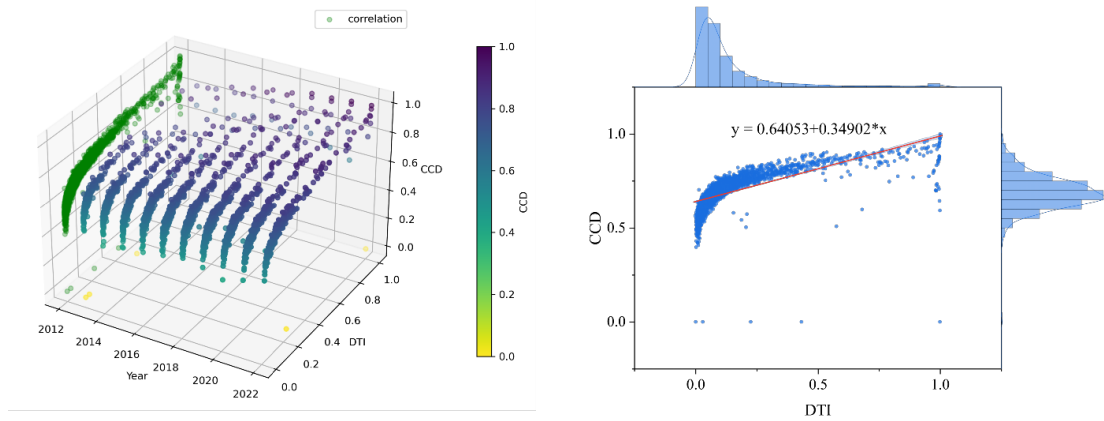


Figure 7 Correlations between *DTI* and *CCD*

5. Empirical analysis

5.1 Impact of digital trade on SEPC

In accordance with the models set in the research design section, we conducted a thorough examination of the potential impact of digital trade on SEPC and adopted both a panel two-way fixed effects model (TWFE) and DDML to estimate treatment effects. The relevant empirical results are given in **Table 4**.

Table 4 Benchmark regression results

Panel A: Treatment effects estimated via the TWFE						
	CCD					
	(1)	(2)	(3)	(4)		
DTI	0.407*** (0.048)	0.247*** (0.064)	0.212*** (0.071)	0.191*** (0.071)		
Controls	No	No	Yes	Yes		
City FE	Yes	Yes	Yes	Yes		
Year FE	No	Yes	No	Yes		
N	2733	2733	2733	2733		
R ²	0.319	0.394	0.425	0.441		
Panel B: Treatment effects estimated via the DDML						
	CCD					
	RF	Lasso	Ridge	GBM	SVM	NN
DTI	0.262*** (0.019)	0.085*** (0.017)	0.091*** (0.019)	0.091*** (0.022)	0.056*** (0.016)	0.085*** (0.031)
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2733	2733	2733	2733	2733	2733

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As shown in Panel A of **Table 4**, the coefficient of digital trade (*DTI*) on SEPC (CCD) is statistically significant at the **1% level** across all columns and remains stable at approximately 0.2 after controlling for some variables, indicating strong economic significance. The regression results suggest that digital trade has a powerful impact on **empowering** regional reductions in carbon and pollutant emissions through synergistic effects, with each one standard deviation increase in *DTI* potentially leading to an improvement of approximately 0.2 standard deviation in the CCD.

In Panel B of **Table 4**, estimation from DDML is consistent with TWFE. To ensure that the results are as reliable as possible, the following machine learning algorithms are used for estimation: random forest (RF), lasso and ridge regression, gradient boosting machine (GBM), support vector machine (SVM), and neural network (NN). The coefficient of interest is still significantly positive at the 1% significance level, indicating that there is a clear promotion effect of digital trade on SEPC, and Hypothesis one is initially supported.

5.2 Robustness test

5.2.1 Addressing omitted factors

To further exclude the impact of omitted factors on the benchmark regression results, this section conducts a robustness test of strengthening fixed effects and augmenting time trends.

First, different provinces have different geographic locations and different focuses on the overall development of the country, and the high administrative level of the province creates obvious interprovincial policy style differentiation; thus, this section adds high-dimensional interaction fixed effects of province and year to the benchmark regression model to control for the yearly characteristics of the province in which the city is located and to prevent the confounding treatment effect of the characteristic factors at the provincial level.

Second, digital trade has continued to show firm growth in recent years, while pollution and carbon reduction have also made great progress, and the existence of the indicator's own development trend may confound the estimated treatment effects and lead to biased causal inference; thus, this section adds a time trend and its interactions with the province FE to make more reliable causal inference. The results are presented in **Table 5**.

Table 5 Results after augmenting high-dimensional fixed effects and the time trend

	CCD					
	Province × Year FE		Time trend		Province FE × Time trend	City FE × Time trend
<i>DTI</i>	0.143*** (0.018)	0.116*** (0.017)	0.252*** (0.064)	0.197*** (0.072)	0.147*** (0.035)	0.324* (0.182)
Controls	No	Yes	No	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	-	-	-	-	Yes	Yes
N	2681	2681	2733	2733	2733	2733
R ²	0.950	0.953	0.387	0.434	0.573	0.744

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As shown in **Table 5**, after strengthening the fixed effects and incorporating time trends into the model, the goodness-of-fit of the regression improved significantly compared with that of the baseline model. The robust coefficients with stronger interpretive power continued to exhibit positive significant signs across various conditions, indicating that the conclusions drawn from the baseline regression remained valid even when accounting for potential omitted variables and other factors that may have influenced the results.

5.2.2 Addressing the sample selection problem

Despite the trimming of extreme values to mitigate potential biases in coefficient estimates, some inherent properties of the sample may still lead to biased estimates.

Four municipalities, such as Beijing and Shanghai, are actually provincial-level administrative regions, participating them in the regression with other cities may result in inflated coefficients; additionally, regression excluding special economic zones, such as Shenzhen and Zhuhai, may be more reflective of most regular cities. The relevant regression results are available in **Table 6**. After excluding municipalities and special economic zones, the coefficients decrease slightly but remain positive and significant, confirming that special cities mildly inflated the baseline results. This finding demonstrates that digital trade promotes SEPC universally, not just in developed cities, reinforcing the robustness of hypothesis one.

Some matching techniques, such as entropy balance matching, could be helpful to construct more comparable treatment and control groups, to gain more reliable estimation. We use the provincial annual mean of *DTI* for classification, and include quadratic and interaction terms of all covariates to capture the nonlinear correlations. **Figure 8** displays the performance of entropy balancing, the standardized difference of covariates before and after matching has significantly decreased, indicating the treatment and control groups are more balanced now. **Table 7** presents the regression results after matching. Despite a slight decrease in the statistical significance of *DTI* coefficients, the economic significance has increased enough to again support the robustness of baseline conclusion.

Table 6 Results after excluding municipalities and special economic zones

	CCD					
	No municipalities		No special economic zones		No both	
<i>DTI</i>	0.177*** (0.021)	0.110*** (0.017)	0.248*** (0.066)	0.193*** (0.072)	0.176*** (0.021)	0.111*** (0.017)
Controls	No	Yes	No	Yes	No	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2693	2693	2693	2693	2653	2653
R ²	0.483	0.557	0.388	0.435	0.476	0.551

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

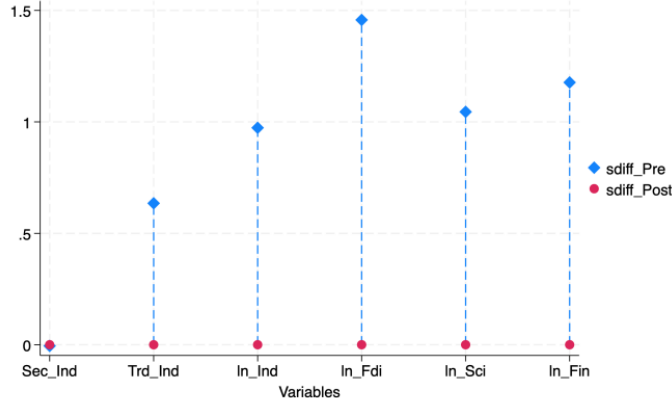


Figure 8 Standardized difference of the variables after entropy balancing

Table 7 Results after entropy balancing

	CCD			
	(1)	(2)	(3)	(4)
<i>DTI</i>	0.537*** (0.185)	0.507* (0.271)	0.532* (0.276)	0.504* (0.267)
Controls	No	No	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes
N	2733	2733	2733	2733
R ²	0.786	0.792	0.797	0.803

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Finally, to ensure the K-fold cross-fitting algorithm for the DDML model doesn't yield different results because of different split folds of the samples, we modified the split folds to 4 and 6 in this section. The results of **Table 8** show that changing the data split fold does not significantly affect the overall performance of the *DTI* coefficients.

Table 8 Validity tests for DDML: change data split folds

	CCD					
	RF	Lasso	Ridge	GBM	SVM	NN
Folds: 4						
<i>DTI</i>	0.247*** (0.017)	0.076*** (0.016)	0.081*** (0.016)	0.114*** (0.019)	0.113*** (0.019)	0.012 (0.036)
Folds: 6						
<i>DTI</i>	0.278*** (0.022)	0.102*** (0.027)	0.103*** (0.024)	0.100*** (0.024)	0.108*** (0.026)	0.099*** (0.021)
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2733	2733	2733	2733	2733	2733

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.2.3 Addressing the subjective weighting and measurement error problem

The dependent variable used in this paper is a composite index composed of multiple subsystems. During the initial construction, we assigned equal weights to the five subsystems for the purpose of balancing and reflecting the coordination of the subsystems. Despite this operation conforming to academic conventions and theoretical requirements, measurement issues arising from weights still

exist. Therefore, we used the entropy weight method to reassign objective weights to the five subsystems of *CCD*, and then used the newly constructed indicator in the regressions. The weights for each subsystem are neither arbitrarily assigned nor equal. The results shown in **Table 9** are consistent with the baseline regression, indicating that our conclusions will not be seriously affected by subjective weighting.

Table 9 Sensitivity test for *CCD*

	<i>CCD_Topsis</i>			
	(1)	(2)	(3)	(4)
<i>DTI</i>	0.378*** (0.020)	0.355*** (0.028)	0.368*** (0.029)	0.353*** (0.030)
Controls	No	Yes	No	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
N	2733	2733	2733	2733
R ²	0.530	0.538	0.537	0.543

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Composite indicators may amplify the measurement error of each variable and have a significant impact on estimation through a “1+1>2” mechanism. To check whether the measurement errors in *DTI* significantly affect the conclusions, we used single indicators from the system for regression, which are e-commerce sales (*EC_Sale*) and software business revenue (*Software*), representing the digitalization of trade methods and trade content respectively. To ensure that measurement errors are not significant at different levels, we also used the geometric mean of the e-commerce sector (*EC_avg*) and the software industry sector (*Software_avg*) in *DTI* for regression respectively. All columns in **Table 10** remain positively significant, and the coefficient sizes also fluctuate within a reasonable range, indicating that measurement errors will not seriously affect our conclusions.

Table 10 Sensitivity test for *DTI*

	<i>CCD</i>			
	(1)	(2)	(3)	(4)
<i>EC_Sale</i>	0.101** (0.043)			
<i>Software</i>		0.112** (0.046)		
<i>EC_avg</i>			0.202** (0.091)	
<i>Software_avg</i>				0.118** (0.048)
Controls	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	2733	2733	2733	2733
R ²	0.430	0.429	0.431	0.448

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.3 Endogeneity processing

5.3.1 Exogenous shock test

Endogeneity may arise from the bidirectional causality between *DTI* and *CCD*. The DID method can effectively address this issue. By exploiting an exogenous policy closely related to digital trade, the causal impact of *DTI* on *CCD* can be identified. Since the policy was not formulated based on cities' *CCD* performance, the risk of reverse causality is substantially reduced.

A quasi-natural experiment was conducted via a DID model based on the pilot policy of Comprehensive Cross-Border E-Commerce Test Zones. Cross-border e-commerce optimizes the trade process by supporting digital means such as online transactions, cross-border payments, and intelligent logistics. It presents a two-sided association with digital trade. Moreover, whether a city should be selected as a pilot city is not affected by the city's SEPC performance, thus satisfying the condition of exogeneity.

The DID regression results show a positive and significant effect across various controls, confirming that the cross-border e-commerce pilot policy enhances regional SEPC efficiency. Since e-commerce embodies trade digitalization, these findings indirectly support the positive role of digital trade in promoting SEPC.

Table 11 *DID analysis of the cross-border e-commerce pilot policy*

	<i>CCD</i>			
	(1)	(2)	(3)	(4)
<i>did</i>	0.042*** (0.003)	0.011*** (0.003)	0.008*** (0.003)	0.007*** (0.003)
Controls	No	No	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes
N	2733	2733	2733	2733
R ²	0.090	0.329	0.381	0.406

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 11 shows that pre-policy trends between treatment and control groups were similar, with significant differences emerging after the policy, validating the DID approach. A mixed placebo test confirms the treatment effect's authenticity, as the true coefficient lies at the distribution's extreme tail. These results further support the positive impact of digital trade on regional SEPC.

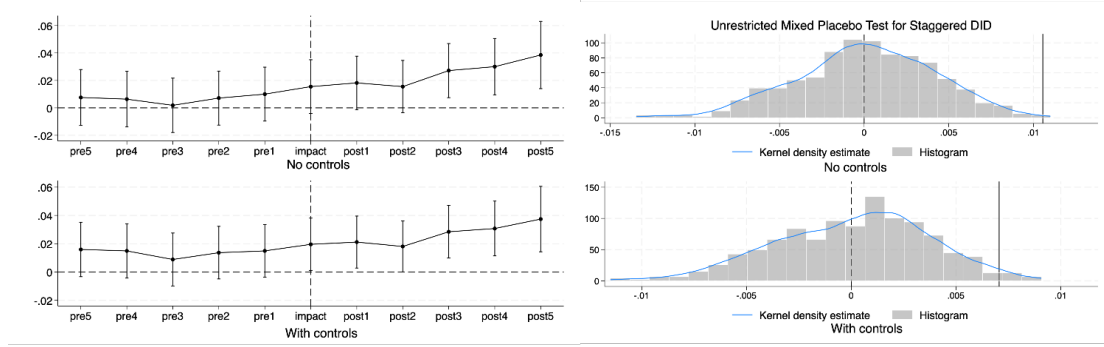


Figure 9 Tests for the rationality of the DID design

5.3.2 Instrumental variable approach

Endogeneity may also result from omitted variables when the error term is correlated with the explanatory variable. In this case, introducing an instrumental variable that is strongly correlated with digital trade but uncorrelated with the error term helps isolate the net effect of digital trade and mitigates bias arising from omitted variable problems.

Following the approach of (Qin et al., 2022), this study selects the spherical distance between each city and Hangzhou as the instrumental variable. Given the significant spatial agglomeration and spillover effects in the development of the internet industry and digital economy, a city's proximity to Hangzhou is expected to influence its potential digital trade capacity, thereby satisfying its relevance. Moreover, geographical distance itself does not directly affect economic growth or environmental pollution, ensuring exogeneity. Since spherical distance constitutes cross-sectional data, this study interacts distance with the previous year's postal service volume to construct panel data for use as an instrumental variable. The results are presented in **Table 12**.

Table 12 Results of the IV method

	CCD			
	(1)	(2)	(3)	(4)
<i>iv.DTI</i>	0.325*** (0.012)	0.082*** (0.028)	0.310*** (0.012)	0.086*** (0.029)
Controls	No	Yes	No	Yes
Year FE	No	No	Yes	Yes
Kleibergen–Paap rk LM statistic	55.504***	52.210***	55.563***	52.753***
Cragg–Donald Wald F statistic	1638.030	803.865	1548.553	817.882
Kleibergen–Paap rk Wald F statistic	139.692	132.300	136.070	134.713
Stock-Yogo 10% critical value			16.380	
N	2441	2441	2441	2441
R ²	0.520	0.775	0.505	0.770

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The instrumental variable passes the KP-LM test, rejecting under-identification, and the F statistics exceed the Stock-Yogo 10% threshold, confirming instrument relevance. After isolating the exogenous component of digital trade, its coefficients remain significantly positive, indicating

a robust promoting effect on regional SEPC even after addressing endogeneity concerns.

The distance to Hangzhou, the acknowledged hub of the digital economy, may correlate with more variables than those incorporated in our baseline model. If the instrument exerts an impact on CCD through channels beyond digital trade, the exclusiveness restriction underlying the IV regression will be violated. Theoretically, whether it is industrial transfer, government intervention after policy learning, or financial radiation effects, they are all covered by the control variables of the benchmark model, so the exclusiveness of instrument can be guaranteed. To provide empirical evidence, we implement a series of exclusiveness tests to verify the validity of the instrument. These tests are grounded in several simple yet robust logics:

First, if the exclusiveness assumption holds, the instrument affects CCD only via digital trade, which means that when DTI is controlled, the coefficient of the instrument should be insignificant and adequately small, whereas the coefficient of DTI remains basically unchanged. This derivation is supported according to results shown in the first two columns in **Table 13**. Second, in the IV regression theory, endogeneity in the independent variables is allowed, which may cause bias to estimations in column (1) and (2). Therefore, we conducted an auxiliary regression of CCD on DTI, and extract the residuals, which captures the component of CCD that's orthogonal to DTI. If the exclusiveness restriction is valid, the IV should exhibit no significant impact on it. This theoretical conjecture is supported by evidences, from column (3). Third, if the instrument were to affect CCD through channels other than digital trade, industrial structure adjustment would emerge as the most plausible confounding pathway. However, the regression results in columns (4) and (5) indicate that when DTI is controlled, the instrument exerts no significant impact on urban industrial structure. This finding thus suggests that the risk of violating the exclusiveness restriction is minimal.

Table 13 *Exclusiveness tests for the instrument*

	<i>CCD</i>	<i>CCD</i>	<i>Res</i>	<i>Sec Ind</i>	<i>Sec Ind</i>
	(1)	(2)	(3)	(4)	(5)
<i>iv</i>		-0.000 (0.000)	-0.000 (0.000)		0.003 (0.004)
<i>DTI</i>	0.349*** (0.041)	0.360*** (0.057)		18.088*** (4.504)	18.478*** (4.858)
<i>N</i>	2733	2441	2441	2733	2441
<i>R</i> ²	0.495	0.523	0.002	0.607	0.550

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Following a series of robustness checks and endogeneity treatments, there is now substantial empirical evidence supporting the validity of Hypothesis 1: digital trade contributes positively to the synergistic development of pollution reduction and carbon emission mitigation.

5.4 Mechanistic analysis of innovation capabilities and factor market distortion

The logic of our causal path inference is as follows: first, we use reduced models to estimate the effect of digital trade on mechanism variables, after confirming that the channels are supported by statistical evidence, we use causal mediation models to decompose the effects. If the mechanism truly exists, the share of the effect transmitted through it should not be too low. The mediating role of regional innovation capability is first confirmed using both *invention* and *utility* to handle different definitions of innovation. According to panels A and B in **Table 14**, both the TWFE and DDML models show a significant positive impact of digital trade on regional innovation capability. This study has already provided substantial theoretical and empirical evidence supporting the positive role of regional innovation in promoting SEPC. Since innovation reduces emissions, drives economic growth, and lowers environmental governance costs, these findings strongly support Hypothesis 2: digital trade enhances innovation, which in turn promotes SEPC.

Table 14 Mediating effects of regional innovation capability and factor market distortion

	TWFE		RF		Lasso		Ridge		GBM	
Panel A: Invention as dep.var.										
DTI	0.160*** (0.044)		0.154*** (0.045)		0.056*** (0.020)		0.053*** (0.019)		0.051*** (0.018)	0.027* (0.016)
R ²	0.164		0.175		-		-		-	-
Panel B: Utility as dep.var.										
DTI	0.438*** (0.062)		0.437*** (0.061)		0.134*** (0.030)		0.143*** (0.031)		0.141*** (0.029)	0.088*** (0.029)
R ²	0.374		0.382		-		-		-	-
Panel C: FMD as dep.var.										
DTI	-0.206*** (0.025)		-0.206*** (0.025)		-0.045*** (0.009)		-0.051*** (0.008)		-0.047*** (0.008)	-0.034*** (0.008)
R ²	0.698		0.699		-		-		-	-
Controls	No		Yes		Yes		Yes		Yes	Yes
City/Year FE	Yes		Yes		Yes		Yes		Yes	Yes
N	2733		2733		2733		2733		2733	2733

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The results in **Table 13** panel C confirm that digital trade reduces factor market distortions, which improves green total factor productivity, value chain upgrading, and industrial optimization—all contributing to SEPC, validating Hypothesis 3.

To address the concern regarding potential measurement errors in the mediating variables, we alternatively use the number of green invention patents classified according to the WIPO system to represent technological innovation, and the labor markdown estimated by GMM estimation and labor adjustment costs correction based on tax investigation data (Yin et al., 2024) to capture factor market distortions. These two new variables are generated from different data source, and from different methods. The results using these alternative indicators in **Table 15** remain largely

unchanged, indicating that potential measurement errors do not fundamentally affect the conclusions of this study.

Table 15 *Robustness tests for measurement error*

	<i>Green Invention</i>			<i>Labor Markdown</i>		
	TWFE	GBM		TWFE	GBM	
<i>DTI</i>	0.356*** (0.074)	0.358*** (0.075)	0.071*** (0.016)	-0.462*** (0.169)	-0.384** (0.181)	-0.332* (0.186)
Controls	No	Yes	Yes	No	Yes	Yes
City/Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2733	2733	2733	1098	1098	1098
R ²	0.276	0.282	-	0.069	0.085	-

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

On the basis of the classical causal mediation model proposed by Imai et al. (2010), this paper decomposes the mediating effects to discuss the relative importance of mechanisms. To ensure estimation accuracy, we employ DDML for bias correction. Similar to Equation (4), we establish the following residualized structural equation model:

$$e_var_{it} = var_{it} - E[var_{it}|X_{it}] \quad (8)$$

$$e_Invention_{it} = \alpha_1 e_DTI_{it} + \sigma_{it} \quad (9)$$

$$e_FMD_{it} = \alpha_2 e_DTI_{it} + \sigma_{it} \quad (10)$$

$$e_CCD_{it} = \beta_1 e_DTI_{it} + \beta_2 e_Invention_{it} + \beta_3 e_FMD_{it} + \sigma_{it} \quad (11)$$

where $E[var_{it}|X_{it}]$ denotes the predicted values of variables estimated via the DDML (random forest) algorithm on the basis of control variables X_{it} , capturing components of variables associated with controls in a flexible, nonlinear form. Subsequently, e_var_{it} represents the residualized portions of variables after adjusting for controls, isolating pure variations unrelated to confounders. By constructing the structural equation model using these residuals, we eliminate confounding biases introduced by control variables and obtain unbiased estimates of mediating effects.

The calculation methods and decomposition results in **Table 16** reveal that enhanced regional innovation capability accounts for approximately 13.33% of the total effect, whereas alleviated factor market distortion contributes 24% to the improvement in the SEPC. **It should be noted that since both the core explanatory and explained variables are composite indices with inherent measurement errors, these specific values are only for comparing the relative importance of the two mechanisms rather than reflecting the absolute magnitude of their respective effects.** As discussed earlier, both innovation enhancement and factor market optimization stem from improvements in data/information circulation, integration, and utilization efficiency. **This implies that the data / information flows via digital trade** directly induce nearly 40% of the observed green benefits,

making it a crucial factor in this digital-green nexus. These findings reaffirm the critical role of data/information circulation in achieving green transformation, and helps to identify the priority of policy intervention for digital trade to exert its green effect.

Table 16 *Mediation effects decomposing*

Effects	Calculation	Value	%
Direct Effect	β_1	0.047	62.759
Total Indirect Effect	$\alpha_1 \times \beta_2 + \alpha_2 \times \beta_3$	0.028	37.185
IE <i>Invention</i>	$\alpha_1 \times \beta_2$	0.010	12.908
IE <i>FMD</i>	$\alpha_2 \times \beta_3$	0.018	24.303
Total Effect	$\beta_1 + \alpha_1 \times \beta_2 + \alpha_2 \times \beta_3$	0.075	100.000

5.5 Heterogeneity analysis

Theoretically, the environmental externalities of digital trade mainly rely on traditional trade; therefore, its impact on SEPC is likely to be influenced by the local trade environment. At the micro level, studies have shown that enterprises receiving government environmental subsidies are more capable of realizing the promoting effect of digitalization on SEPC, and the alleviation of corporate financing constraints is an important mechanism behind this (Chong et al., 2024). Meanwhile, enterprises located in free trade zones enjoy financing advantages brought by additional policy support, and face more stringent environmental regulations simultaneously. Therefore, the impact of digital trade on SEPC should be stronger within free trade zones. At the organizational level, authoritative literature has confirmed the network structure of international trade. Enterprises located at key nodes of the network are more able to enter the international market and will also use network nodes to search for new partners (Chaney, 2014). The improvement of the business environment helps to enhance enterprises' ability to participate in the global value chain, thereby improving regional resource allocation efficiency (Zou & Lei, 2023), which is the most important mechanism found in this paper through which digital trade enhances SEPC. Therefore, the trade liberation policies of free trade zones can enable enterprises to be more deeply embedded in the international network, which is beneficial to the development of digital trade. Additionally, studies have shown that integrated management in industrial parks helps to enhance synergistic effects, thereby contributing to SEPC (Wang et al., 2024). In summary, the favorable trade environment and collaborative management in free trade zones can amplify the impact of digital trade on SEPC.

Industrial structure changes and transfers are important mechanisms in many digital economy literatures. For example, the digital economy effectively improves local SEPC level through industrial upgrading, while the transferred energy-intensive and pollution-intensive industries have

a significant negative impact on the SEPC of neighboring regions (Li et al., 2024). Structural studies have found that the agglomeration of manufacturing industries weakens the carbon emission reduction effect of digital trade, while the agglomeration of production service industries can strengthen this effect (Wang et al., 2023). The agglomeration of digital industries can improve local energy efficiency (Xu et al., 2024), thereby affecting SEPC efficiency (Zhao et al., 2024). It can be seen that whether as an intermediary or an external input shock, the industrial structure has an important impact on regional digital trade development and SEPC. Existing literature indicates that industrial structure upgrading, mainly characterized by the transfer of manufacturing industries and the agglomeration of service industries, can expand the impact of digitalization on environmental governance.

Cities with larger populations often have stronger urban vitality (Lan et al., 2020), which means a larger market, a favorable condition for the development of digital trade. At the same time, the concentration of a large population in big cities directly means a larger scale of energy consumption and pushes up carbon emissions through the “industrial structure effect” and “traffic effect” (Yi et al., 2022). The concentration of population in cities also creates a large demand for real estate construction, and existing literature has proven that real estate investment in most cities at the current stage significantly deteriorates air quality (Chen & Lee, 2020). That is to say, big cities are more conducive to the development of digital trade, while also having larger-scale pollutant and greenhouse gas emissions. Therefore, the impact of digital trade on SEPC should be stronger in bigger cities.

Table 17 provides empirical evidences supporting the above derivations. These findings highlight how contextual factors—trade policies, industrial maturity, and city size—modulate the effectiveness of digital trade in driving environmental sustainability, offering nuanced insights for targeted policy design.

Table 17 Results of heterogeneity tests

	CCD					
	FTZs	Non-FTZs	AIS	Non-AIS	Large	Nonlarge
<i>DTI</i>	0.208*** (0.076)	0.141 (0.107)	0.138*** (0.041)	0.060 (0.081)	0.227** (0.090)	0.086 (0.053)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
City/Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2077	656	1366	1367	1361	1372
R ²	0.458	0.411	0.435	0.357	0.412	0.513

Note: Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.6 Empirical analysis of the threshold effect

Digital trade's expansion can **affect** both industrial growth and higher energy consumption, potentially leading to pollution and carbon emissions. This suggests an optimal level where its environmental benefits peak before diminishing. To explore this nonlinear relationship, a threshold effect model using *DTI* as the threshold variable identifies critical turning points, offering insights for policies that maximize digital trade's sustainability impact.

Figure 10 presents the likelihood ratio (LR) across candidate threshold values. The LR curve surpasses the critical value, confirming the presence of a threshold effect. The threshold estimate is identified at 0.139, where the LR statistic attains its minimum value below the critical line, marking the optimal threshold point for model differentiation.

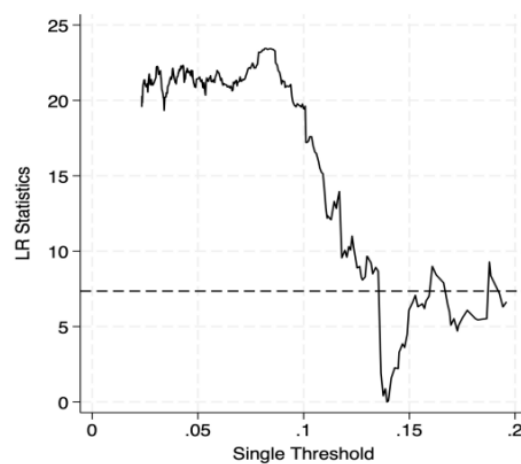


Figure 10 *LR statistics*

According to the results in **Table 18**, the threshold effect model, tested with 500 bootstrap replications, reveals a significant single threshold at $DTI = 0.139$. When *DTI* exceeds this level, the marginal effect of digital trade on SEPC decreases by 29.27%, indicating a nonlinear relationship. Beyond this threshold, the environmental benefits of digital trade are partially offset by increased pollution and carbon emissions. These results suggest that an optimal level of digital trade exists, and cities exceeding this threshold will require complementary environmental policies, such as stricter emission regulations, to balance trade growth with sustainability.

Table 18 *Results of the panel threshold model*

<i>Panel A: Results of threshold regression</i>			
	Coefficients		Robust Std. err.
$DTI \cdot I(DTI \leq 0.139)$	0.328***		0.0271
$DTI \cdot I(DTI > 0.139)$	0.232***		0.0139
Controls	Yes		
City FE	Yes		
Year FE	Yes		
N	2550		
R ²	0.477		

<i>Panel B: Results of threshold effect test</i>			
Threshold Var.	Threshold Value	F Value	P Value
<i>DTI</i>	0.139	23.523	0.048

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

This finding is consistent with the classic “digital rebound” effect, suggesting that digital transformation and the new digital economy itself have a tendency toward high energy consumption and high emissions. Drawing on the China's experiences, in the early stage of the digital transformation of trade, it was mainly carried out spontaneously and scattered by trade enterprises, and mainly focused on the application of digital technologies to optimize traditional trade processes (Li et al., 2020). At this stage, there was barely no large-scale new investment in advanced digital infrastructure such as data centers and computing centers. Therefore, digital trade at this stage could better promote SEPC through virtualization, technology diffusion, and market optimization. In the more advanced stage, trade in the digital industry begins to expand. On the one hand, this directly leads to massive constructions and productions in the ICT and broader digital service industries, resulting in the regional agglomeration of supporting manufactures (Wang, 2023). It is also likely to attract the establishment of free trade zones and digital industrial parks through policy channels (Ma et al., 2019), which in turn brings higher energy consumption and carbon emissions.

However, since it involves regional economic growth and environmental governance simultaneously, more consideration should be given to emission efficiency rather than absolute scale. Focusing on the green development of digital trade itself, and thereby further stimulating the sustainable role of digital trade in promoting clean economic growth is a major policy recommendation of this paper. Meanwhile, the reason why Chinese cities' digital trade has not shown a “backfire” effect in the early stage is that digital trade mainly relies on existing trade industries and trade structures. For other aspects of digitalization, this effect may be very obvious. Therefore, this paper suggests that the digital performance in different economic fields may have differentiated characteristics that are of great theoretical and practical significance.

6. Conclusions and policy implications

6.1 Research conclusions

This paper quantifies the detailed development of China's urban digital trade by establishing a prefectural indicator system. Data reveals that while China's urban digital trade is growing rapidly, significant regional disparities dominated by the Matthew effect persist, with the Pearl River Delta

emerging as the top-performing urban cluster. Building upon the CCDM, this study constructs an SEPC assessment incorporating economic benefits. The regional disparity of SEPC is milder, reflecting relatively balanced improvements in environmental governance efficiency.

The results uncover a stable and robust relationship between digital trade and SEPC. Digital trade promotes urban SEPC by virtualizing trade goods and production processes, enhancing trade efficiency to conserve resources, and facilitating virtual agglomeration of digital industries. Technology diffusion mechanisms from classic international trade theory remains a primary channel here. Additionally, cross-border information and data flows integrate international trade networks, alleviating factor market distortions, thus a more efficient markets simultaneously enhance the effectiveness of environmental governance and sustainable economic growth. The environmental context of a city's trade activities significantly influences this effect: a more open trade atmosphere, more favorable trade conditions, and larger trade markets all amplify the environmental efficacy of digital trade. Finally, the emerging digital rebound effect is also evident here, with the promotion of SEPC by digital trade being offset by an energy-consumption rebound of approximately 29.27% after crossing a threshold.

Overall, these findings suggest that digital trade functions as a double-edged driver of urban green development: it improves environmental efficiency primarily through market and technology channels, while also generating new environmental pressures at more advanced stages of development.

6.2 Theoretical contributions

This paper endeavors to shift the research perspective from mechanical environmental externality to the efficiency of environment-economy coupled coordination. It provides quantitative evidence for qualitative literatures on digital trade's environmental benefits, and offers more micro empirical details for the digital environmental externality theory. Theoretically, the findings extend classic international trade theories into the digital context by showing how virtualization and network integration reshape technology diffusion and market optimization, thereby influencing environmental governance efficiency. Moreover, by identifying a significant rebound effect, this study enriches the understanding of the Digital Paradox, demonstrating that digitalization does not monotonically show environmental performance but follows a stage-dependent trajectory. This dynamic perspective offers a more nuanced theoretical explanation for the heterogeneous

environmental externalities of digitization observed across regions.

6.3 Practical value

According to the conclusions, the virtualization of traded goods and trade processes, as well as the integration of international trade networks based on cross-border information flows, are the underlying logic through which digital trade promotes SEPC. Therefore, policymakers should proactively adapt to this mechanism, use targeted subsidy policies to promote R&D of digital goods and the application of digital technologies, and actively seek innovations in international trade coordination mechanisms under the impact of digitalization. For example, establishing a cross-border information sharing consensus framework, multi-agent negotiations to formulate inclusive digital service trade guidelines, and collaborative opening up in data and other digital goods.

Technology diffusion based on international trade networks is one of the main mechanisms through which digital trade affects SEPC. Therefore, the government should encourage various agents to absorb technological diffusion in trade from multiple aspects. For enterprises, a recognition mechanism for advanced learning enterprises can be established to commend and offer preferential treatment to enterprises that actively introduce and learn new technologies and new management schemes. For governments, they should endeavor to increase technology contracts, take technology transfer as a factor in trade negotiations, absorb China's experience of “market for technology”.

Market optimization is the most important channel through which digital trade transmits positive environmental externalities. Policymakers should actively use data opening and information sharing to reduce distortions in markets. In a macro perspective, further reducing barriers to factor mobility between different markets is meaningful. Specifically, governments or large enterprises should be encouraged to lead in establishing digital trade consultation and market information service platforms, further reduce factor misallocation caused by insufficient information, and achieve higher trade efficiency.

The expansion of trade in digital industries will lead to an increase in overall energy consumption, thereby exerting pressure on environmental governance. Therefore, a sustainable digital development plan must be established. Hierarchical energy management systems could be useful to control the production and transfer of digital pollution. Digital equipment recycling should be incorporated into the circular economy framework, and strict carbon footprint accounting standards

for new digital industries must be developed in time. Also, environmental protection assessments should be carried out first to ensure that digital development matches the environmental carrying capacity of cities.

6.4 Research limitations and prospects

Certain limitations inevitably exist in this study. First, the indicator system is constrained by the inherent flaws of the entropy weighting method, which may assign lower weights to variables with imbalanced data distributions yet rich economic implications, failing to fully reflect their true importance. Second, in order to incorporate major components of digital trade, this study adopts synthesized indicator as core explanatory variable, such practice introduces unsolvable measurement errors. Third, constrained by data availability and the need to avoid explosive complexity in the index structure, critical variables such as cross-border data flows and digital platforms are not incorporated, which may lead to biased estimates of related coefficients. Meanwhile, the high requirements of the indicator system for data availability and quality further result in difficulties in covering the annual data of all prefecture-level cities across China. Lastly, the analysis focuses on Chinese cities, and the conclusions may not be directly generalizable to other institutional contexts.

For future research, these flaws could be addressed by optimizing the weighting method, expanding the scope of key variables under the premise of controlling measurement errors, and incorporating multi-source authoritative statistical data to build a more comprehensive and interpretable evaluation framework with broader sample coverage. Future research could incorporate enterprises or cross-country data to explore micro mechanisms and comparative institutional effects.

Appendix

1. Calculation of factor market distortion (*FMD*)

The essence of factor market distortion is the deviation between what a factor should obtain and what it actually achieves, which is manifested in the ratio of the marginal factor output to the factor price. The Translog production function finds the partial derivatives of labor (L) and capital (K) for Equation (1) and introduces the marginal outputs of labor and capital.

$$\ln Y_{it} = \lambda_0 + \lambda_1 \ln L_{it} + \lambda_2 \ln K_{it} + \frac{1}{2} \lambda_3 (\ln L_{it})^2 + \frac{1}{2} \lambda_4 (\ln K_{it})^2 + \lambda_5 \ln K_{it} \ln L_{it} \quad (\text{a1})$$

$$MPL_{it} = (\lambda_1 + \lambda_3 \ln L_{it} + \lambda_5 \ln K_{it}) Y_{it} / L_{it} \quad (\text{a2})$$

$$MPK_{it} = (\lambda_2 + \lambda_4 \ln K_{it} + \lambda_5 \ln L_{it}) Y_{it} / K_{it} \quad (\text{a3})$$

At this point, the degree of factor distortion for L and K can be obtained by dividing MPL_{it} and MPK_{it} by the interest rate (r) and the wage (w), respectively, which are factor prices:

$$T_{L,it} = MPL_{it} / w_{it} \quad (\text{a4})$$

$$T_{K,it} = MPK_{it} / r_{it} \quad (\text{a5})$$

Finally, the *FMD* for the whole market is the geometric average of the L and K distortions:

$$FMD_{it} = \sqrt{T_{L,it} \times T_{K,it}} \quad (\text{a6})$$

In Equations (1-6), Y represents the regional GDP of the year, L denotes the number of employed individuals at the end of the year, K is the total fixed asset investment of the year, w is the average annual salary per worker, and capital price r is the quoted loan market interest rate (LPR) for the current year issued by the Chinese National Interbank Funding Center.

CRedit authorship contribution statement

Fengxiu Zhou: Writing – original draft, Formal analysis. **Lei Li:** Writing – review and editing, Data curation. **Chien-Chiang Lee:** Writing – review and editing, Visualization, Project administration, Conceptualization.

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